

SEMIH ESKI

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SKILLS

Mechanical: SolidWorks, NX, Ansys (FEA), GD&T, Tolerance Stack-up, DFMA, FMEA/Root-Cause Analysis
Electrical / SW: Raspberry Pi, ESP32, PCBA Design, Altium, Instrumentation, Actuators, Python, MATLAB, C++
Manufacturing: CNC, Injection Molding, Composite Fabrication, Additive Manufacturing, Sheet Metal, Lathe, Mill

EXPERIENCE

Mechatronics Engineering Intern

May 2026 – Present

Electrans Technologies Ltd.

Oakville, ON

- Designed & built an automated electro-pneumatic press using SolidWorks to insert 600,000 compression limiters into production parts, cutting cycle time 83%, eliminating operator injury and part-damage risks; produced GD&T drawings for fabrication
- Designed a portable metrology fixture that builds a full datum reference frame off the trailer kingpin, resolving installation hole positions to 0.2 mm across 3,000+ in-service trailers spanning 30"/36" kingpin setbacks
- Designed & built a custom spring-loaded retractable pin system with counterbore datum locating for insertion alignment, designed for high-cycle reliability through PFMEA, worst-case tolerance stack-up, and fatigue-life sizing of components
- Designed & built a real-time IP67 electromechanical truck-boom-angle measuring system using a hall-effect rotary sensor driven by a four-bar linkage; validated sensor accuracy and repeatability to 0.1° via test setup
- Designed & prototyped custom tooling, fixtures and protective enclosures across the production line, accelerating manufacturing cycle time 40% while improving operator safety and eliminating part damage

Mechanical Engineering Intern

Aug 2024 – Dec 2024

Equans Services

Montreal, QC

- Performed complex WC & RSS tolerance stack-up analyses to verify conveyor belt alignment within ± 5 mm, across a 6 m long conveyor
- Performed 30+ quality control inspections on production-line equipment, diagnosing & replacing 6+ faulty parts, restoring operation

EXTRACURRICULAR

Co-Founder & Mechanical Team Lead

Dec 2025 – Present

Waterloo Automation Collective

Waterloo, ON

- Co-founded a 10+ member machine design student team; directing the full mechanical design & launch of a motorized autonomous 15-ton injection molding machine with 100 MPa injection capability and 3 cubic inch shot volume
- Owning the design & build of a 15-ton five-point toggle clamping mechanism; synthesized linkage kinematics in MATLAB, validated via FEA in ANSYS; releasing GD&T drawings backed by a 50+ variable worst-case 3D stack-up
- Designed and built an automatic ejection unit with a 10×10 pin grid, accommodating manufacturing tolerances, varying mold sizes and geometries, while maintaining constant stroke and clamping tonnage without retooling

Mechanical Project Manager & Designer

Aug 2025 – Aug 2026

Waterloo Aerial Robotics Group (WARG)

Waterloo, ON

- Led clean-sheet design of an STOL UAV empennage in SolidWorks, driven by configuration trade studies; root-caused structural failures and automated design for stability-and-control through a MATLAB toolchain
- Engineered a foldable drone-arm mount in SolidWorks, achieving 80% mass reduction; verified durability via a 300-cycle fatigue test
- Manufactured R&D composite airframes via split-mold tooling and multi-method layups to maximize strength-to-weight ratio

PROJECTS

CubeSat Structural FEA & Vibration Analysis

- Performed non-linear quasi-static FEA on UW Orbital's CubeSat in ANSYS to simulate launch conditions, validating structural integrity
- Cut down computation time by 14 hours, by achieving 81% high-quality hex-dominant mesh via strategic partitioning of the geometry
- Conducted modal and random vibration (PSD G) FEA on the pretensioned chassis by solving for modes until >90% effective mass in each axis, finding low-amplification IMU mounting zones by mapping 3-sigma random-vibration displacement/stress

Autonomous Object-Picking Vehicle

- Achieved 95% real-time detection, ± 5 cm distance precision, implementing machine vision on Raspberry Pi using a custom Python/OpenCV vision pipeline for detecting ping pong balls and range estimation; debugging real-time data via iterative tests
- Developed C++ PID navigation using IR & ToF sensors and a low-latency UART between RPi & ESP32 to execute real-time commands

EDUCATION

University of Waterloo – GPA 3.9/4.0

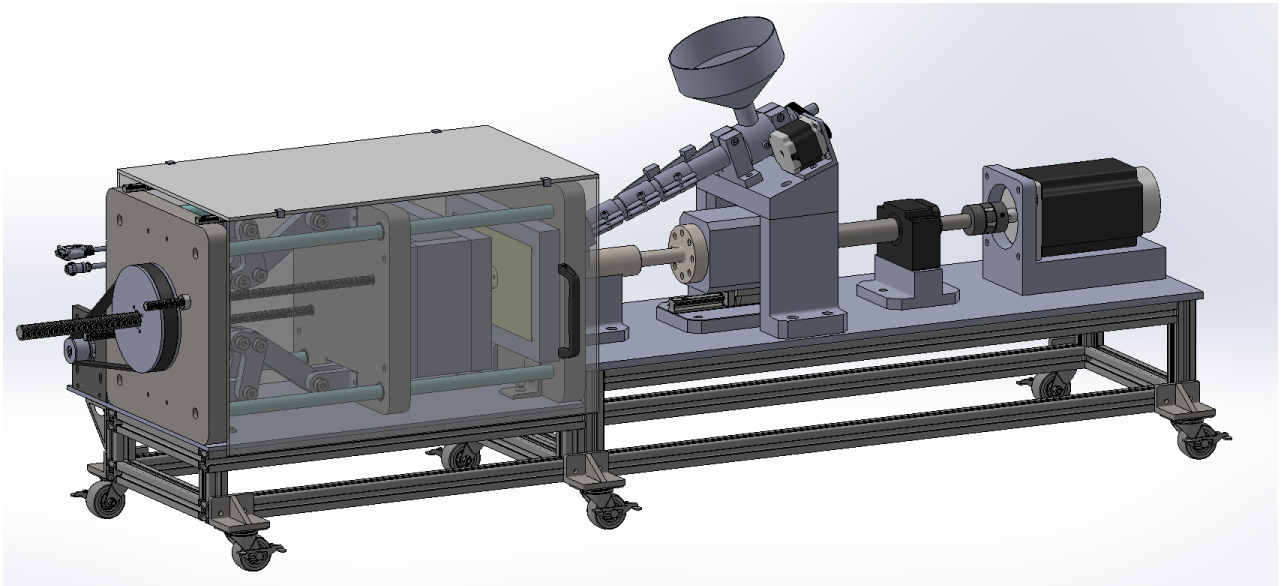
Aug 2025 – Present

Candidate for BAsC - Honours Mechanical Engineering, Co-op

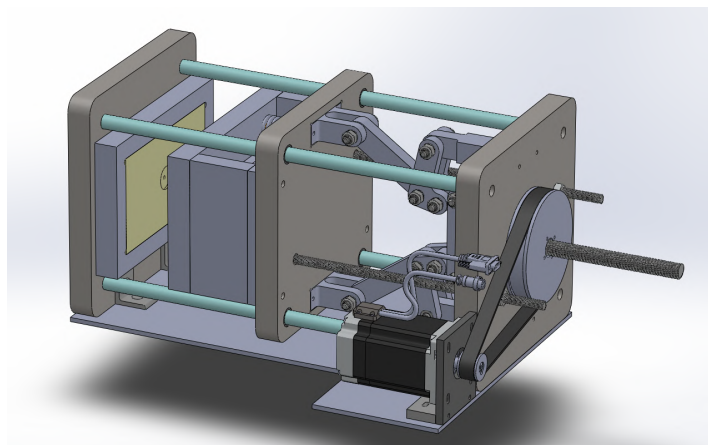
Waterloo, ON

Motorized V-Line Injection Molding Machine @ WAC - [Link to Media](#)

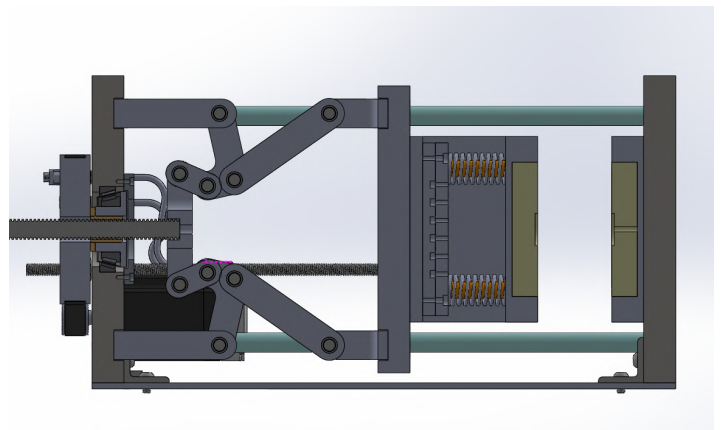
- Co-founded a 10+ member machine design student team & leading the mechanical design of a motorized V-line injection molding machine (15-ton clamping, 100MPa injection)
- Sized screws, bearings, and structure against buckling, thread stripping, yielding, fracture, fatigue, thermal expansion, and brinelling; designed guides against binding and stick-slip, and isolated components from cycle vibration
- Synthesized toggle linkage kinematics in MATLAB to meet tonnage, stroke, and mechanical-advantage targets
- Producing a 50+ variable 3D worst-case tolerance stack-up and GD&T drawings for fabrication



Injection Molder Full CAD



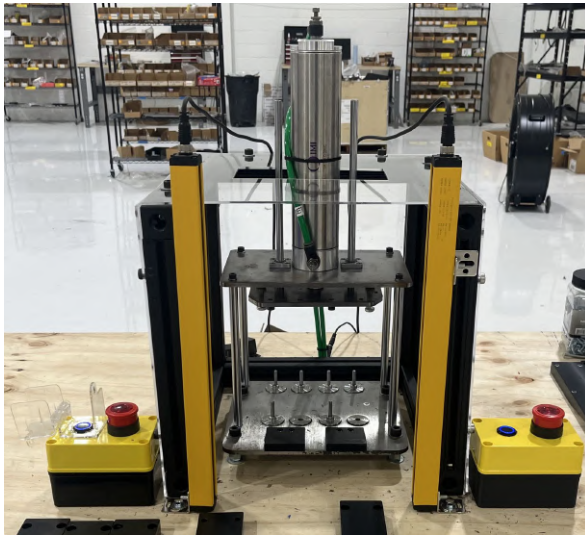
Clamping, Ejection, and Mold (CEM) Units View



CEM Section View

Automated Electro-pneumatic Precision Insertion Machine @ Electrans Technologies - [Link to Media](#)

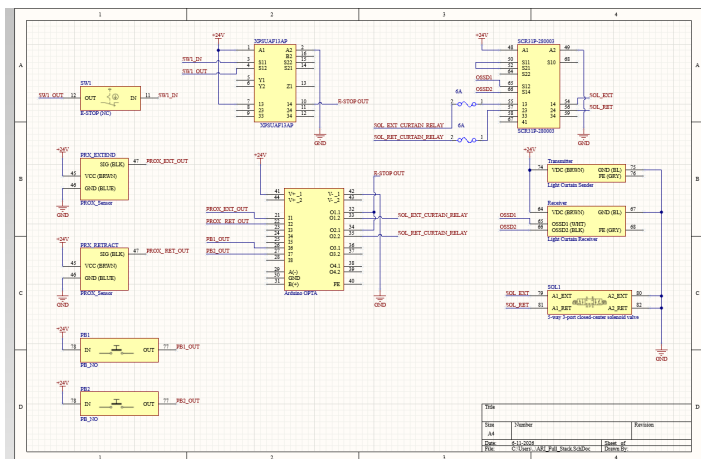
- Designed and built an automated electro-pneumatic press to insert 600,000 compression limiters, cutting cycle time 83% and eliminating operator injury and part-damage risk
- Engineered the safety architecture: anti-tie-down two-hand control, light curtain with safety relays, and a 5/3 closed-center valve for automatic retraction
- Designed the full electrical schematic in Altium and pneumatic circuit; released GD&T drawings backed by worst-case tolerance stack-ups
- Owned the full product launch: PFMEA, subsystem testing, process flow, work instructions, SOPs, and installation manuals.



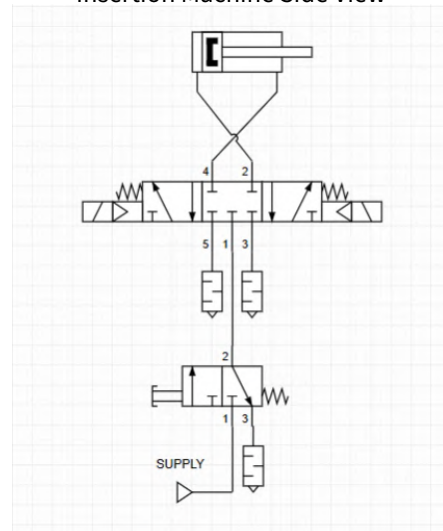
Insertion Machine Front View



Insertion Machine Side View



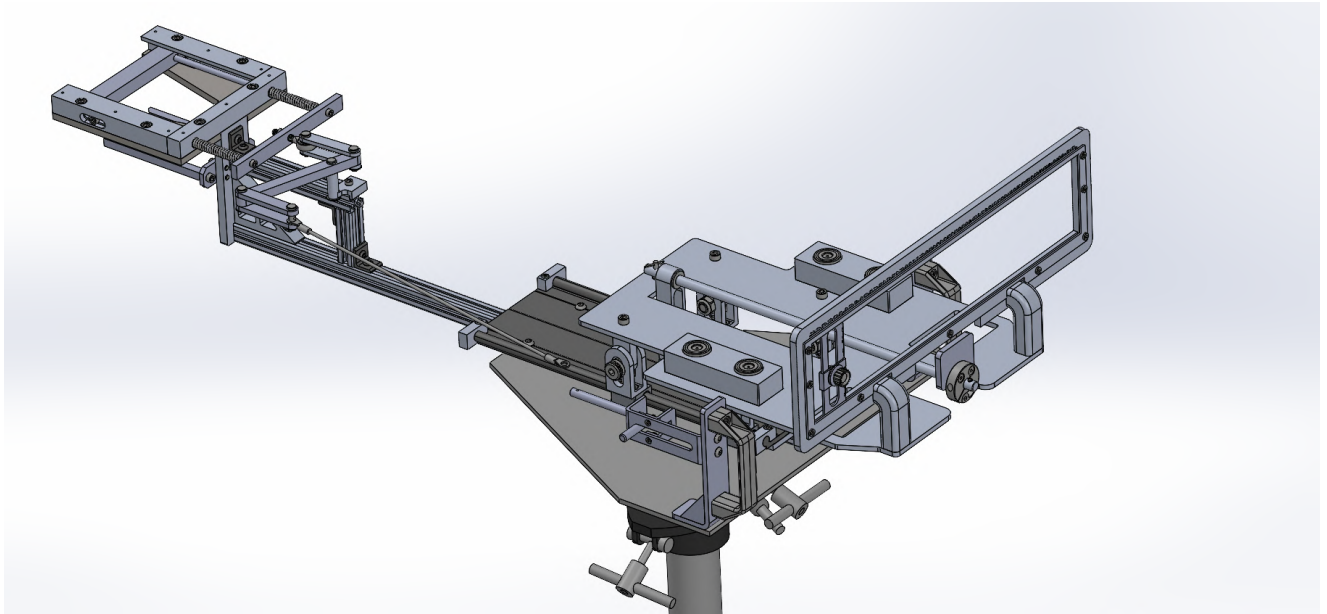
Electrical Schematic in Altium



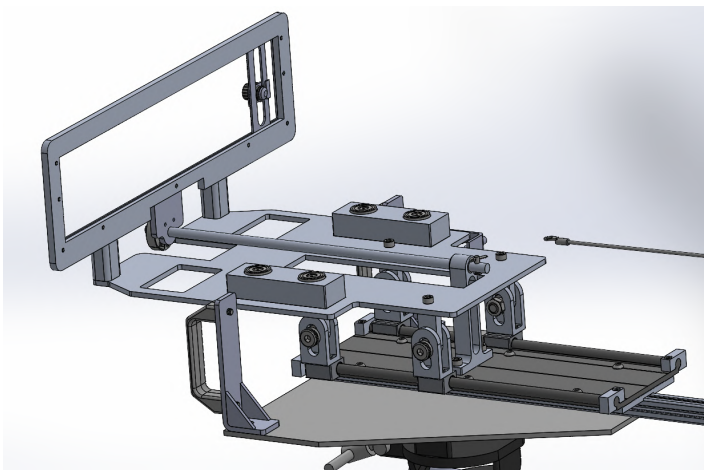
Pneumatic Diagram

Self-Datuming 3-axis Coordinate Metrology Fixture @ Electrans Technologies - [Link to Media](#)

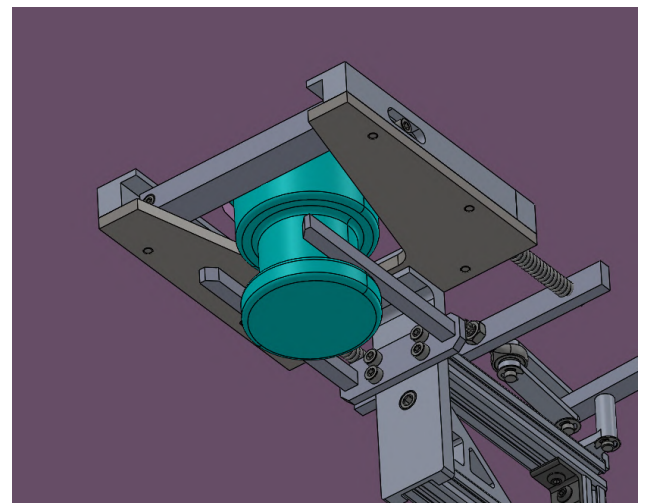
- Designed a portable metrology fixture that builds a full datum reference frame off a trailer kingpin, measuring hole positions to 0.2 mm in X & Y across 3,000+ trailers spanning 30"/36" kingpin setbacks
- Applied Euclidean compass bisection against a V-block seat to repeatably datum the fixture off the trailer kingpin centerline despite highly variable trailer geometry and tolerances
- Designed a custom XY multistage measuring system to locate holes and read coordinates down to 1/32" resolution
- Conducted material trade studies to design against galvanic corrosion, surface abrasion and wear, and fatigue failure, while accounting for ergonomics
- Released GD&T drawings backed by a 17-variable worst-case tolerance stack-up from kingpin seat through the XY multistage measurement system



Metrology Fixture ISO



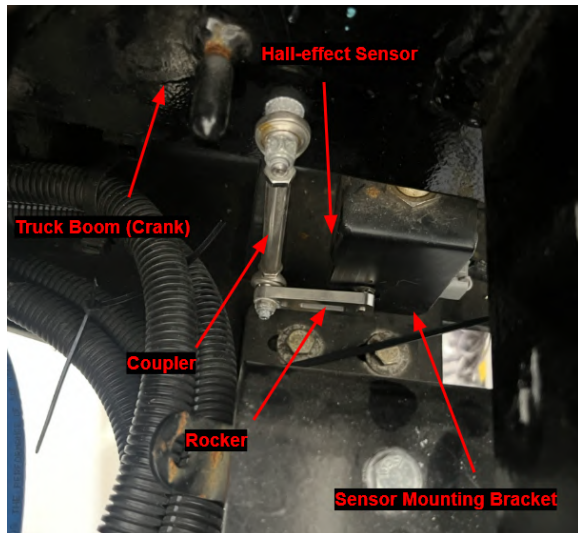
Y-Datum Carriage Plate Subassembly



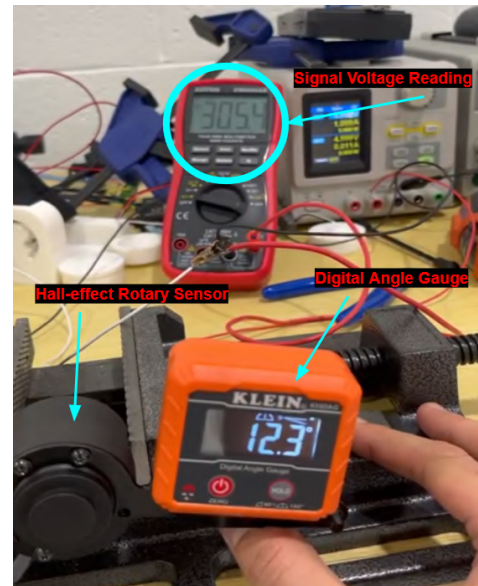
X-Datum V-Block Subassembly

Electromechanical Truck Boom Angle Sensor @ Electrans Technologies - [Link to Media](#)

- Designed and built an IP67 boom-angle measurement system using a hall-effect rotary sensor driven by a four-bar linkage; compared draw wire vs ToF vs IMU vs hall effect sensors
- Synthesized the linkage for a 1:1 input-to-output angle ratio, preserving linearity across the full boom sweep
- Validated accuracy and repeatability to 0.1° by testing sensor signal voltage with a digital angle gauge



Assembled & Mounted System



Sensor Validation

Production Line Tooling & Fixture Design @ Electrans Technologies - [Link to Media](#)

- Designed & prototyped custom tooling, fixtures and protective enclosures across the production line, accelerating manufacturing cycle time 40% while improving operator safety and eliminating part damage
- Conducted time studies as part of tooling configuration trade analysis, and used DFMA & poka-yoke principles to optimize for ergonomics, safety, and efficiency



Example 1 - Assembly Station Fixture



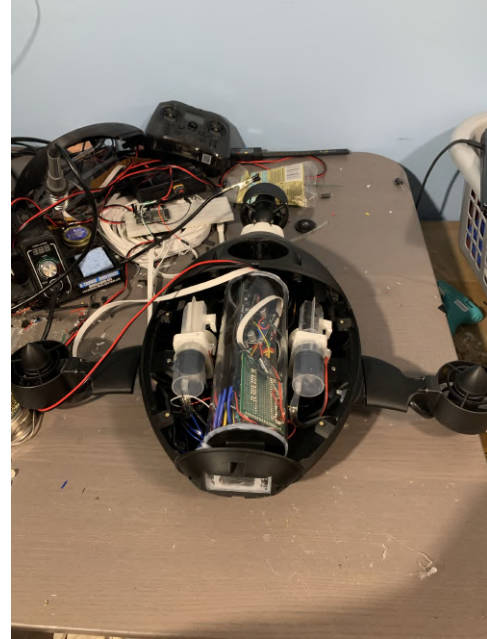
Example 2 - Assembly Station Fixture

Amphibious Underwater Drone - [Link to Media](#)

- Designed an ROV via top-down master modeling and surface modeling in SolidWorks, driving parametric updates across a 15+ part assembly while holding G2 curvature continuity for better hydrodynamic efficiency
- Refined FDM parameters and applied acetone vapor smoothing to seal microporosity, achieving a watertight 550 g ABS frame at IPX7 with 15% neutral buoyancy margin
- Integrated a thrust-vectoring propulsion system and instrumented the ROV for submersion testing



Fully Built Frame



Top Lid Removed



Frame Top View



Frame Front View

Fixed-wing UAV (Helios) & Competition Hex Drone (Valkyrie) @ WARG

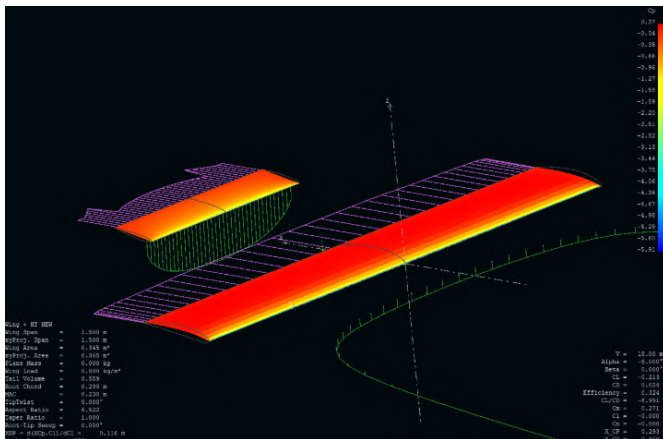
- Led the full aerodynamic and mechanical design of the Helios STOL empennage, from configuration trade study through flight test; automated the stability-and-control design in a MATLAB/XFLR5 toolchain
- Actuated both elevator halves synchronously from a single servo via a coupled torque tube, saving weight and eliminating differential-deflection trim error; root-caused and redesigned in-service tail failures
- Managed mission-critical tasks on Valkyrie, including hardware mounting architecture for the fire-extinguishing task and Li-ion battery pack holders sized for flight loads and CG limits



Helios - Maiden Flight



Helios - Ground Testing



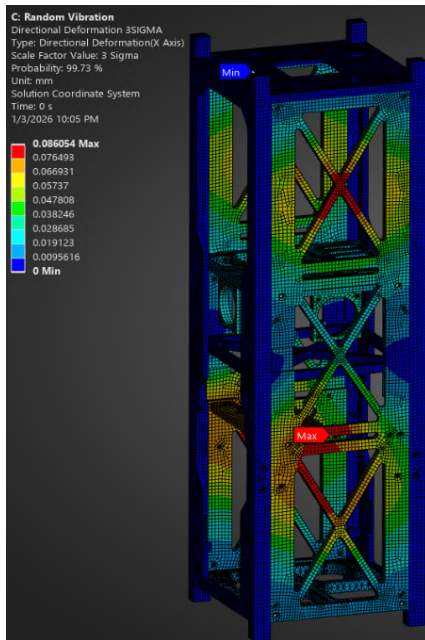
Helios - Aero Stability Simulation on XFLR5



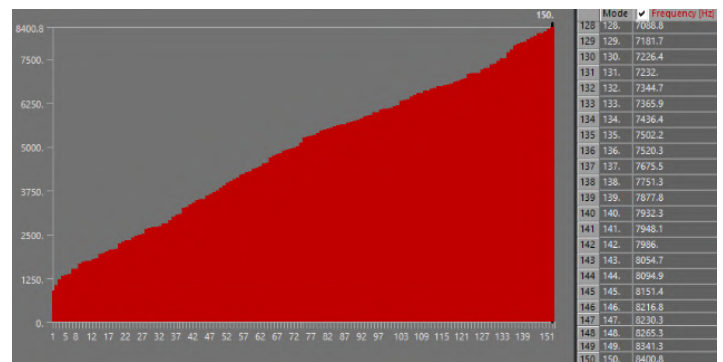
Valkyrie - WARG Competition Drone

Non-Linear Quasi-Static, Modal, and Random Vibration FEA of CubeSat @ UW Orbital

- Solved the random vibration (PSD G) study to $1.5 \times$ PSD max frequency and $\geq 90\%$ effective mass per axis, mapping 3-sigma displacement to find low-amplification IMU mounting zones
- Cut computation time 14 hours by partitioning geometry for $\geq 81\%$ hex-dominant mesh with local refinement at stress gradients



Total Deformation (X Dir.) under Random Vibration



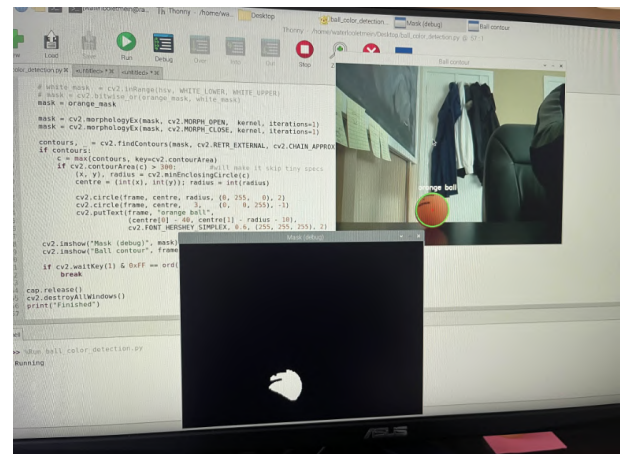
Chassis Resonant Frequency Modes

Autonomous Computer Vision Vehicle

- Built a custom Python/OpenCV pipeline on a Raspberry Pi for ping-pong ball detection and range estimation, hitting 95% real-time detection and ± 5 cm precision
- Developed C++ PID navigation on IR and ToF feedback, with a low-latency UART link between the Pi and ESP32 for real-time drive commands



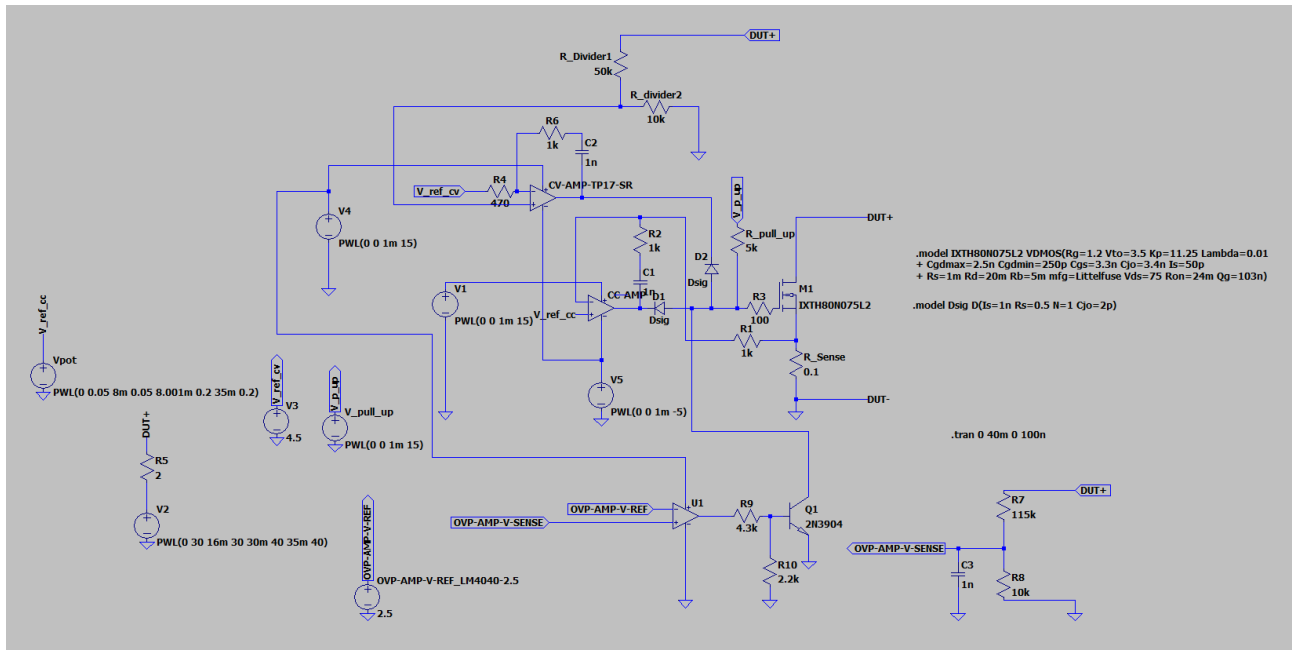
Vehicle in Action



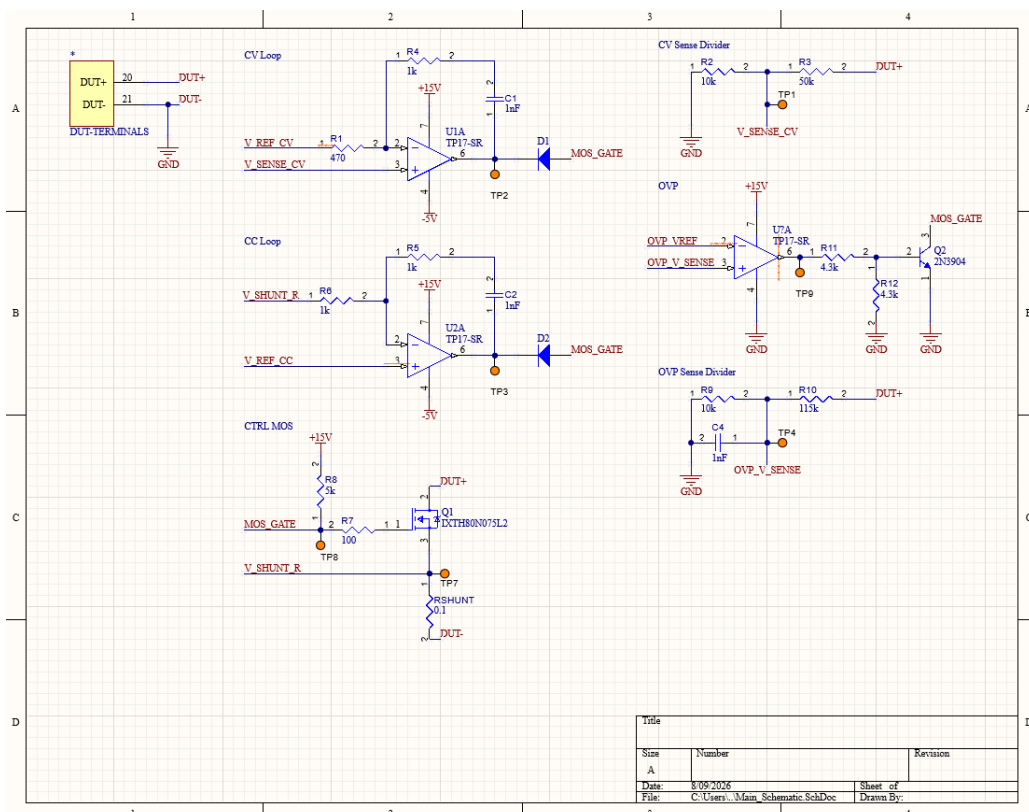
Python OpenCV Script Successful Detection of Ping Pong Ball

Programmable Electronic Load - Link to Media

- Designing a 50 W / 30 V / 5 A DC electronic load with a fully analog control loop, supporting constant voltage, constant current, and constant power modes; equipped with overvoltage and overcurrent protection
- Implemented mode arbitration via a diode-OR network of transconductance amplifiers, so the most restrictive loop takes control of the pass element without contention
- Ran transient analyses in LTSpice to verify loop stability, mode-handoff behavior, and step response



LTSpice Transient Analysis Circuit



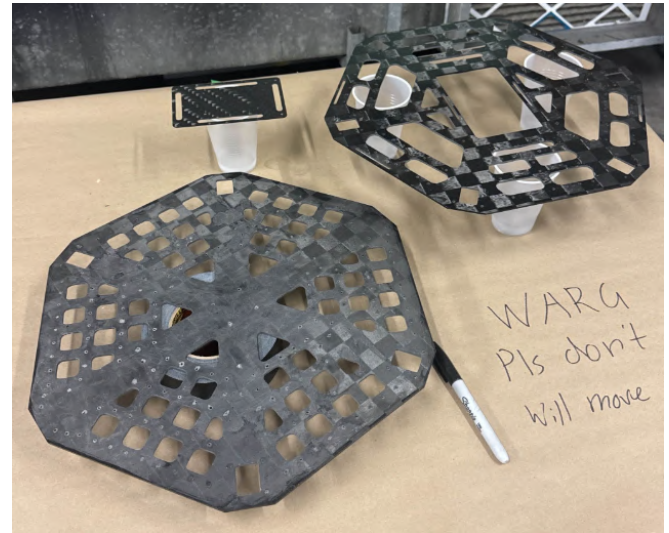
Altium Schematic & PCB Layout (In Progress)

Composite Manufacturing @ WARG

- Fabricated high-strength composite UAV airframe components via split-mold tooling, wet layup, and foam-core layup
- Executed a 7-ply quasi-isotropic wet layup with manual consolidation and vacuum bagging; post-processed cured plates by sanding and epoxy-sealing edges to prevent delamination



Composite Wet Layup Process



Wet Layup Post-Processed



Split-Mold Vacuum Bagged



Split Molded Airfoil vs Foam Core Layup